Week 10
Networking and JDBC
Program 28
Write a Java program demonstrating TCP client-server communication.
Program 1: TCPServer.java

```
import java.net.*;
import java.io.*;

class TCPServer {
   public static void main(String[] args) throws IOException {
      ServerSocket server = new ServerSocket(5000);
      System.out.println("Server is waiting for client...");

      Socket socket = server.accept();
      System.out.println("Client connected.");

      BufferedReader in = new BufferedReader(
            new InputStreamReader(socket.getInputStream()));

      PrintWriter out = new PrintWriter(
            socket.getOutputStream(), true);

      String message = in.readLine();
      System.out.println("Client: " + message);

      out.println("Hello from Server");

      socket.close();
      server.close();
   }
}
```

Program 2: TCPClient.java

```
import java.net.*;
import java.io.*;

class TCPClient {
   public static void main(String[] args) throws IOException {
      Socket socket = new Socket("localhost", 5000);

      BufferedReader in = new BufferedReader(
            new InputStreamReader(socket.getInputStream()));

      PrintWriter out = new PrintWriter(
            socket.getOutputStream(), true);

      out.println("Hello from Client");
```

```
        String response = in.readLine();
        System.out.println("Server: " + response);

        socket.close();
    }
}
```

Output:
Server console:
Server is waiting for client...
Client connected.
Client: Hello from Client

Client console:
Server: Hello from Server

Program 29
Write a Java program to establish a JDBC connection with a MySQL database.

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.SQLException;

class JDBCConnection {
    public static void main(String[] args) {
        String url = "jdbc:mysql://localhost:3306/college";
        String user = "root";
        String password = "";

        try {
            Connection con = DriverManager.getConnection(
                    url, user, password);

            System.out.println("Database connected successfully.");

            con.close();
        } catch (SQLException e) {
            System.out.println("Connection failed: "
                    + e.getMessage());
        }
    }
}
```

OUTPUT:
Database connected successfully.

Program 30
Write a Java program to perform Insert, Update, Delete, and Select operations using JDBC.

```
import java.sql.*;

class JDBCOperations {
   public static void main(String[] args) {
      String url = "jdbc:mysql://localhost:3306/college";
      String user = "root";
      String password = "";

      try (Connection con = DriverManager.getConnection(
            url, user, password);
          Statement st = con.createStatement()) {

         st.executeUpdate(
            "CREATE TABLE IF NOT EXISTS Student (" +
            "id INT PRIMARY KEY, name VARCHAR(50), age INT)");

         // Insert
         st.executeUpdate(
            "INSERT INTO Student VALUES (1, 'Rahul', 20)");
         System.out.println("Record inserted.");

         // Update
         st.executeUpdate(
            "UPDATE Student SET age = 21 WHERE id = 1");
         System.out.println("Record updated.");

         // Select
         ResultSet rs = st.executeQuery(
            "SELECT * FROM Student");

         System.out.println("Student Records:");
         while (rs.next()) {
            System.out.println(
               "ID: " + rs.getInt("id") +
               ", Name: " + rs.getString("name") +
               ", Age: " + rs.getInt("age"));
         }
         rs.close();

         // Delete
         st.executeUpdate(
            "DELETE FROM Student WHERE id = 1");
         System.out.println("Record deleted.");

      } catch (SQLException e) {
         System.out.println("Database error: "
               + e.getMessage());
      }
```

```
    }
}
```

OUTPUT:

```
Record inserted.
Record updated.
Student Records:
ID: 1, Name: Rahul, Age: 21
Record deleted.
```